

Stimuli Data Generation

Signal Composition

The test signals were designed to cover specific verification cases:

- **Left Channel:** A mix of a 5 Hz sine wave (Amplitude 15,000) and random Gaussian noise. This tests the datapath's handling of dynamic, non-monotonic signals.
- **Right Channel:** A linear ramp signal sweeping from -20,000 to +20,000. This provides a deterministic slope to verify *Zero-Crossing* detection and *Gain* linearity.

Data Formatting

The script simulates the behavior of a 16-bit ADC by clipping all values to the signed integer range $[-32768, 32767]$. The data is exported to a text file (`stimuli.txt`) with two columns (Left, Right) for the VHDL testbench to read.

Python Generation Script

```
import numpy as np

N = 100                      # number of samples
t = np.linspace(0, 1, N)

# Create different types of audio test signals
sine = 15000 * np.sin(2*np.pi * 5 * t)      # 5 Hz sine
noise = 2000 * np.random.randn(N)           # small random noise
ramp = np.linspace(-20000, 20000, N)         # increasing ramp

# Combine signals
left = sine + noise
right = ramp

# Clip to 16-bit signed (ADC Simulation)
left = np.clip(left, -32768, 32767).astype(int)
right = np.clip(right, -32768, 32767).astype(int)

# Write to text file for VHDL Testbench
with open("stimuli.txt", "w") as f:
    for L, R in zip(left, right):
        f.write(f"{L} {R}\n")
```

Table 1: Operation Control Vector Encoding (3-bit)

Code (Binary)	Operation Symbol	Function
000	$Y' = L + R$	Addition (Mix)
001	$Y' = L - R$	Subtraction
010	$Y' = R - L$	Reverse Subtraction
011	$Y' = L$	Pass Left Channel
100	$Y' = R$	Pass Right Channel
101	$Y' = -L$	Invert Left Channel
110	$Y' = -R$	Invert Right Channel
111	N/A	Reserved / Output Zero

Table 2: Saturation Mode Control (1-bit)

Code	Mode	Description
0	Unsaturated	Standard overflow (wrap-around) behavior.
1	Saturated	Clipping behavior (clamps to max/min values).

Table 3: Gain Mode Control Encoding (2-bit)

Code	Factor	Operation	Implementation
00	$1.0\times$	Unity Gain	No Shift
01	$2.0\times$	Amplification	Logical Shift Left (<code>sll 1</code>)
10	$0.5\times$	Attenuation	Arithmetic Shift Right (<code>sra 1</code>)
11	$0.25\times$	Strong Attenuation	Arithmetic Shift Right (<code>sra 2</code>)

Table 4: IOM Board Switch and LED Assignments

IOM Component	Left Channel (Conn 1)	Right Channel (Conn 4)
<i>Switches (Inputs)</i>		
Switch 0	Global Enable (<code>enable</code>)	<i>Unused / NC</i>
Switch 1	Op Mode Bit 0 (<code>op_mode_L[0]</code>)	Op Mode Bit 0 (<code>op_mode_R[0]</code>)
Switch 2	Op Mode Bit 1 (<code>op_mode_L[1]</code>)	Op Mode Bit 1 (<code>op_mode_R[1]</code>)
Switch 3	Op Mode Bit 2 (<code>op_mode_L[2]</code>)	Op Mode Bit 2 (<code>op_mode_R[2]</code>)
Switch 4	Saturation Enable (<code>sat_mode_L</code>)	Saturation Enable (<code>sat_mode_R</code>)
Switch 5	Gain Bit 0 (<code>gain_mode_L[0]</code>)	Gain Bit 0 (<code>gain_mode_R[0]</code>)
Switch 6	Gain Bit 1 (<code>gain_mode_L[1]</code>)	Gain Bit 1 (<code>gain_mode_R[1]</code>)
<i>LEDs (Outputs)</i>		
LED 0	Zero Cross Rising (<code>L_H_L</code>)	Zero Cross Rising (<code>L_H_R</code>)
LED 1	Zero Cross Falling (<code>H_L_L</code>)	Zero Cross Falling (<code>H_L_R</code>)
LED 2	Positive Saturation (<code>P_Sat_L</code>)	Positive Saturation (<code>P_Sat_R</code>)
LED 3	Negative Saturation (<code>N_Sat_L</code>)	Negative Saturation (<code>N_Sat_R</code>)